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0178/01

October/November 2016

1 hour

Additional Materials: Geometrical Instruments
Tracing Paper (optional)

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

If working is needed for any question it must be shown below that question.

ELECTRONIC CALCULATORS MUST NOT BE USED IN THIS PAPER.

The number of marks is given in brackets [] at the end of each question or part question.
The total of the marks for this paper is 60.

This document consists of 11 printed pages and 1 blank page.



1 (a) Evaluate.

(i) $3\frac{1}{6} - 1\frac{1}{2}$

Answer (a)(i) [2]

(ii) $1\frac{3}{5} \div 3$

Answer (a)(ii) [2]

(iii) 6.3×0.07

Answer (a)(iii) [2]

(b) Arrange the numbers in order of size, starting with the smallest.

0.66 $\frac{4}{7}$ $\frac{2}{3}$ 0.67 0.667

Answer (b) [2]

- 2 (a) Express 9% as a decimal.

Answer (a) [1]

- (b) Evaluate $3 - 5 \times 2 + 8$.

Answer (b) [1]

- 3 (a) Evaluate.

(i) $3^0 + 3 - 1$

Answer (a)(i) [1]

(ii) $49^{\frac{1}{2}}$

Answer (a)(ii) [1]

- (b) Find the value of n , given that $\left(\frac{1}{2}\right)^n = 4$.

Answer (b) $n =$ [1]



- 4 Matšelisio has a cake.
She gives $\frac{2}{7}$ of it to Teboho and $\frac{1}{6}$ of it to Thato.

Find the fraction of the cake left, giving your answer in its simplest form.

Answer [2]

- 5 In a class, the ratio of boys to girls is 3 : 5.
There are 15 boys in the class.

Find

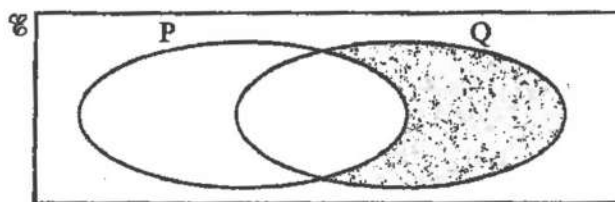
- (a) the number of students in the class,

Answer (a) [2]

- (b) the percentage of girls.

Answer (b) % [1]

- 6 The Venn diagram shows sets P and Q.



- (a) Use set notation to describe the shaded region.

Answer (a) [1]

- (b) $U = \{1, 2, 3, \dots, 15\}$

$P = \{\text{multiples of 3}\}$

$Q = \{\text{prime numbers}\}$

List the elements of

- (i) $P \cap Q$,

Answer (b)(i) [1]

- (ii) $(P \cup Q)'$.

Answer (b)(ii) ... [1]

- 7 (a) Given that $A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}$, find $2A - B$.

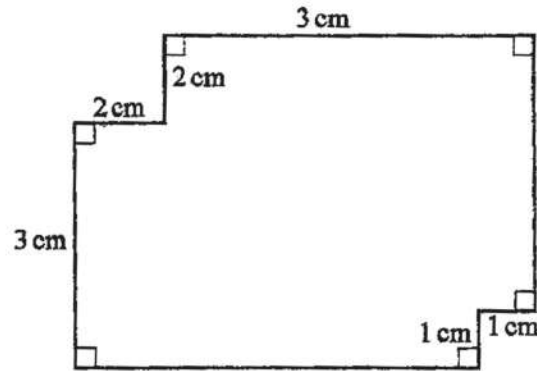
Answer (a) [2]

- (b) If $\begin{pmatrix} k \\ 3 \end{pmatrix} + \begin{pmatrix} -2 \\ p \end{pmatrix} = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$, find the value of k and the value of p .

Answer (b) $k =$

$p =$ [2]





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SCALE

Calculate

(a) the perimeter,

Answer (a) cm [2]

(b) the area.

Answer (b) cm^2 [2]

- 9 There are 5×10^4 people watching a football match in a stadium.
 3.4×10^3 of these are children.

(a) Calculate the number of adults in the stadium.
 Express your answer in standard form.

Answer (a) [2]

(b) The entrance fee into the stadium is M15.00 for adults and M5.00 for children.

Calculate the total amount of money collected

Answer (b) M = [2]

10 $c = x(a + b)$

(a) Evaluate c when $x = 3$, $a = 5$ and $b = -7$.

Answer (a) [2]

(b) Make b the subject of the formula.

Answer (b) $b =$ [2]



11 (a) Factorise.

(i) $a^2b + 6b^2a$

Answer (a)(i) ... [2]

(ii) $b^2 - 36$

Answer (a)(ii) ... [2]

(b) Simplify.

$$x - \frac{1}{3} + x + \frac{1}{2}$$

Answer (b) ... [2]

(c) Solve the simultaneous equations.

$$3x + 2y = 9$$

$$2x - 4y = 22$$

Answer (c) $x =$...

$y =$.. [3]

12 The minimum temperatures ($^{\circ}\text{C}$) in Mokhotlong for 10 consecutive days are

3 1 0 -2 4 1 2 1 -1 -3.

Find

(a) the range,

Answer (a) $^{\circ}\text{C}$ [1]

(b) the median,

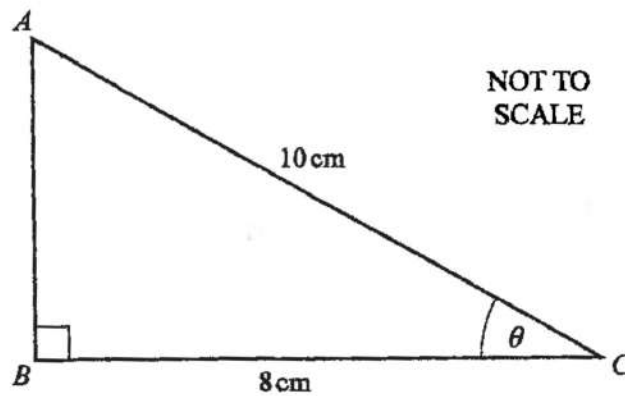
Answer (b) $^{\circ}\text{C}$ [2]

(c) the mode.

Answer (c) $^{\circ}\text{C}$ [1]



- 13 In the diagram, angle $ABC = 90^\circ$, $BC = 8$ cm and $AC = 10$ cm.



- (a) Calculate AB .

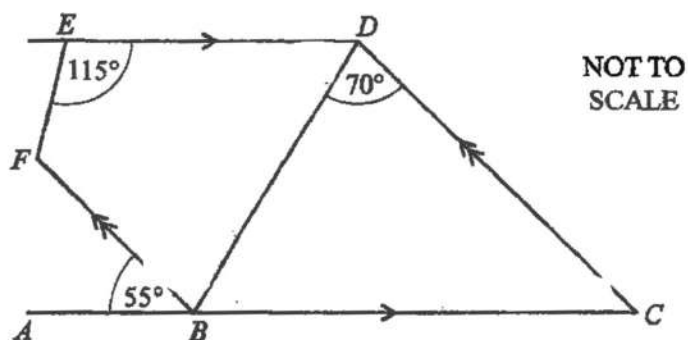
Answer (a) cm [2]

- (b) Write the value of $\cos \theta$ in its simplest form.

Answer (b) [2]

- 14 In the diagram, the straight line ABC is parallel to ED and BF is parallel to CD .

Angle $ABF = 55^\circ$, angle $DEF = 115^\circ$ and angle $BDC = 70^\circ$.



- (a) Find angle CBD .

Answer (a) [1]

- (b) Give the name of triangle BCD .

Answer (b) ... [1]

- (c) State, with a reason, the value of

- (i) angle BDE ,

Answer (c)(i) reason [2]

- (ii) angle BFE .

Answer(c)(ii) reason [2]

